Implementation of Fuzzy Inference System

PROGRAM:

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt
temperature = np.arrange(0,41,1)
fan_speed = np.arrange(0,11,1)
temp_low = fuzz.trimf(temperature,[0,0,20])
temp_medium = fuzz.trimf(temperature,[10,20,30])
temp_high = fuzz.trimf(temperature,[20,30,40])
plt.figure(figsize=(10,6))
plt.subplot(2,1,1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Functions")
plt.xlabel("Temperature (°C)")
plt.tight_layout()
plt.show()
temperature_input = 28
temp_low_level = fuzz.interp_membership
temp_medium_level = temp_medium, temperature_input)
temp_high_level =(temperature, temp_high, temp_input)
fan_low_level = temp_low_level.
fan_medium_level = temp_medium_level.
fan_high_level = temp_high_level.
fan_output = fuzz.defuzz(fan_speed, fan_low*fan_low_level
        + fan_medium_level + fan_high_level,'centroid')
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```

OUTPUT:

Temperature: 28°C

Fuzzified fan speed: 6.00